Hiring Process Analytics

By Gayatri Palkar

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Data Preprocessing

I. Handling Missing Data: Decide on the best strategy to handle them.

- Missing values in this data exists in forms of blank data cells or hyphens '-' in place of data
- I found a blank cell in Offered Salary column, when i checked the minimum and maximum value of column, the difference was too big, so I decided to omit complete row of data, instead of taking the mean, median or mode of data, as there is too much variety and range
- The other columns where I found missing data were categorical data points, hence even there i decided to drop the complete record.

	A	B	C	D	E	F	G
1	application_id	Interview Taken on	Status	event_name	Department	Post Name	Offered Salary
0	114584	07-05-2014 08:08	Rejected	Male	Sales Department	i7	
70							

	A	B	C	D	E	F	G
1	application_id	Interview Taken on	Status	event_name	Department	Post Name	Offered Salary
7	289907	01-05-2014 07:44	Hired	Male	Sales Department	-	85914
7169							

- I found another column with missing values, ***event_name*** column, there should have been two options here: Male and Female, but due to user confidentiality, few applicants had selected “Don’t want to say” in in Gender and there were also few hyphenated cells, so I decided to convert the hyphen data cells to “Don’t want to say”, as it will reduce the number of variables at least by 1. As number of “Don’t want to say” is very high, I decided to not drop all the cells (there were 408 entries with this)

	A	B	C	D	E	F	G
1	application_id	Interview Taken on	Status	event_name	Department	Post Name	Offered Salar
16	195323	09-05-2014 12:48	Hired	-	Service Department	i7	81757
18	742283	02-05-2014 08:11	Rejected	-	Service Department	i5	100
22	834101	01-05-2014 22:53	Rejected	Don’t want to say	Operations Department	i1	82510
32	896164	01-05-2014 17:21	Hired	Don’t want to say	Production Department	i7	37947
93	269204	08-05-2014 07:57	Hired	Don’t want to say	Purchase Department	i5	43093
108	702304	12-05-2014 08:24	Rejected	Don’t want to say	Operations Department	i6	45539
118	814453	13-05-2014 09:19	Hired	Don’t want to say	Operations Department	i7	68515
125	521234	13-05-2014 11:03	Hired	Don’t want to say	Operations Department	i4	44582
134	385418	16-05-2014 11:58	Rejected	Don’t want to say	Service Department	b9	3527
157	505365	16-05-2014 18:06	Hired	Don’t want to say	Human Resource Department	i5	79702
158	132154	16-05-2014 18:07	Rejected	Don’t want to say	Human Resource Department	i5	82599
161	813519	29-05-2014 09:19	Rejected	Don’t want to say	Production Department	i7	83585
163	830139	19-05-2014 00:05	Rejected	Don’t want to say	Operations Department	i4	2103
164	45162	19-05-2014 00:05	Hired	Don’t want to say	Operations Department	i4	1212
182	716333	20-05-2014 17:03	Hired	Don’t want to say	General Management	c-10	71461

Steps

- In both the cases: I selected the blanks and deleted the rows manually using::

1. Select data using Ctrl+A
2. F5
3. Select ->Blanks
4. Delete rows
5. Same with Hyphenated Data rows

	A	B	C	D	E	F	G
1	application_id	Interview Taken on	Status	event_name	Department	Post Name	Offered Salar
16	195323	09Don't want to say05[	Hired	Don't want to say	Service Department	i7	81757
18	742283	02Don't want to say05[	Rejected	Don't want to say	Service Department	i5	100
22	834101	01Don't want to say05[	Rejected	Don't want to say	Operations Department	i1	82510
32	896164	01Don't want to say05[	Hired	Don't want to say	Production Department	i7	37947
93	269204	08Don't want to say05[	Hired	Don't want to say	Purchase Department	i5	43093
108	702304	12Don't want to say05[	Rejected	Don't want to say	Operations Department	i6	45539
118	814453	13Don't want to say05[	Hired	Don't want to say	Operations Department	i7	68515
125	521234	13Don't want to say05[	Hired	Don't want to say	Operations Department	i4	44582
134	385418	16Don't want to say05[	Rejected	Don't want to say	Service Department	b9	3527

II. Clubbing Columns: Do so to simplify your analysis.

- I took a careful look at the data and decided that the ***Department*** and ***Post Name*** as both of them provide similar information, I'll explain how.
- One Department will have many applicants applying for different posts for the overall development of the project as a whole end to end.
- And further on combining these two will help us understand how many people applied for a specific post in a specific department. And will make our analysis easier.

Steps

- For creating a new column by concatenating 2 columns I used the formula:

  =CONCATENATE(E1,"_",F1)

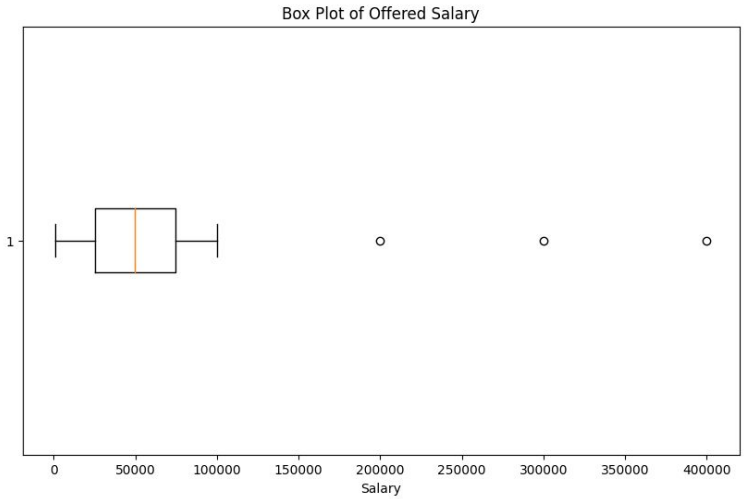
- This joined the 2 columns, then I dragged down the cursor to apply it to the complete column

- Hid the other 2 columns from view

	A	B	C	D	G	H
1	application_id	Interview Taken on	Status	event_name	Offered Salary	Department_PostName
2	12073	14-08-2014 16:09	Hired	Don’t want to say	54951	Finance Department_i5
3	12277	16-08-2014 06:55	Hired	Don’t want to say	22823	Service Department_i7
4	12289	03-06-2014 17:43	Hired	Male	65697	Marketing Department_c5
5	12307	13-06-2014 14:36	Rejected	Female	82707	Sales Department_c5
6	12320	02-06-2014 13:10	Rejected	Male	56727	Production Department_c8
7	12430	28-08-2014 18:40	Hired	Male	65673	Sales Department_b9
8	12713	28-05-2014 14:28	Hired	Female	53360	Operations Department_c9
9	12882	25-06-2014 13:00	Hired	Don’t want to say	43137	Operations Department_i1
10	12887	29-06-2014 17:47	Hired	Male	13201	Human Resource Department_i6
11	12977	21-08-2014 16:38	Hired	Male	39477	Operations Department_i6
12	13205	26-05-2014 15:32	Hired	Male	3045	Operations Department_c5
13	13367	20-05-2014 18:35	Hired	Female	85560	Operations Department_c9
14	13485	09-06-2014 12:22	Rejected	Female	6563	Sales Department_c8

III. Outlier Detection: Check for outliers in the dataset that may skew your analysis.

- To detect outliers, we will have to plot a box plot.
- Since I don't have “**Boxes and Whiskers**” feature in my Excel, I decided to plot a boxplot using python
- We can see in the image that there are 3 outliers, so now to detect them using Excel I will use Quartile function



Python code for box plot

```python
import pandas as pd
import matplotlib.pyplot as plt
```

[1]

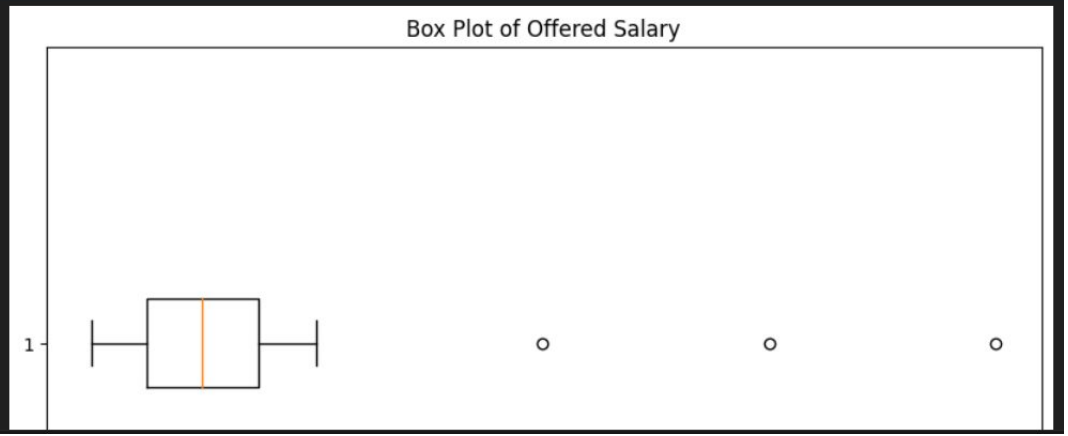
```python
data = pd.read_excel('Statistics_project04_preprocessed1.xlsx')
```

[6]

```python
salary = data['Offered Salary']

plt.figure(figsize=(10, 6))# plotting a box plot to visualize outliers
plt.boxplot(salary.dropna(), vert=False)
plt.title('Box Plot of Offered Salary')
plt.xlabel('Salary')
plt.show()
```

[7]



Outliers using Excel

- To detect the outliers, I will start by finding the Min and Max values from ***OfferedSalary*** column
- Then I will find the Median, Q1 and Q3 using QUARTILE function:

  =QUARTILE(G:G,1) and *=QUARTILE(G:G,3)* and *=MEDIAN(G2:G7152)*

- Then find the Interquartile range(IQR) using formula: *=M22-M20* i.e Q3 - Q1

Min Offered Salary	800	Outliers	
1st Quartile Q1	25516.5	IQR = Q3-Q1	48871
Median	49625	Upper bound Q3+(1.5*IQR)	147694
3rd Quartile Q3	74387.5	Lower bound Q1-(1.5*IQR)	-47790
Max Offered Salary	400000		

Outliers using Excel

- Then I calculated the Upper bound and Lower bound : *=M22+(1.5*P20)* and *=M20-(1.5*P20)*
- using formula: Upper bound = Q3+(1.5*IQR) and Lower bound = Q1-(1.5*IQR)
- Then I used a OR function and created a new column called Outliers and compare the data value with upper and lower bound as : *=OR(G2>P21,G2<P22)*
- This gave me a new column with True and False values for outliers
- I filtered data on ***True*** values and got below output

	A	B	C	D	G	H	I
1	application_id	Interview Taken on	Status	event_name	Offered Salary	Department_PostName	Outlier
4604	649039	07-05-2014 10:48	Hired	Female	200000	Service Department_b9	TRUE
5621	795330	15-06-2014 09:45	Hired	Female	400000	General Management_i4	TRUE
6243	874368	21-07-2014 15:39	Hired	Male	300000	General Management_i7	TRUE
7153							
7154							
7155							
7156							

IV. Removing Outliers/Anomalies: Decide on the best strategy to handle outliers.

- The outliers are very far away from out data and hence will not provide a meaningful insight, so I will remove the outliers completely from the data
- Delete outliers and clean up data
- The Outliers are now deleted, I will also delete or hide the new column ***Outliers*** as our job for cleaning the data is done.
- I will now start with further analysis

V. Data Summary: Summarize the findings.

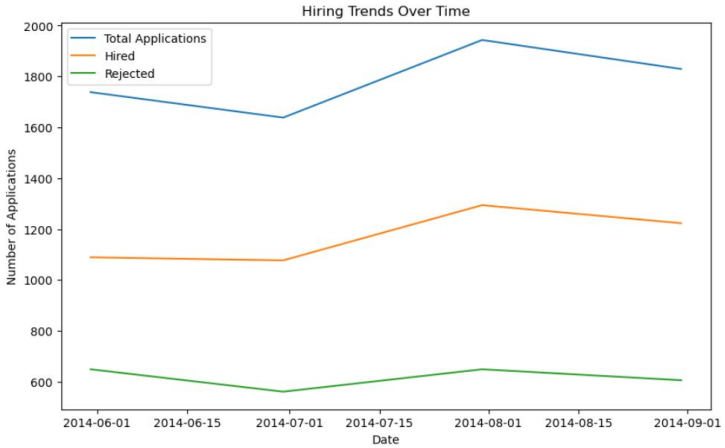
- I found the Statistical measures like Mean, Media, Mode from the ***OfferedSalary*** column, the only numerical column in data
- Also found the maximum and minimum salary offered
- Have also checked the data and removed any Outliers, as it might change further analysis.
- We can see that the mean and median of salaries is similar approx. but salary of 69K is appearing most in the data. But this information is not very valuable. We can further divide these values for department and post-wise Salary
- Let us get into few visualizations using Excel

Mean (Average)	49986.07342
Median	49625
Mode	69583
Max Offered Salary	400000
Min Offered Salary	800

Creating visualizations to better understand the data.

We can look at Hiring trends over time:

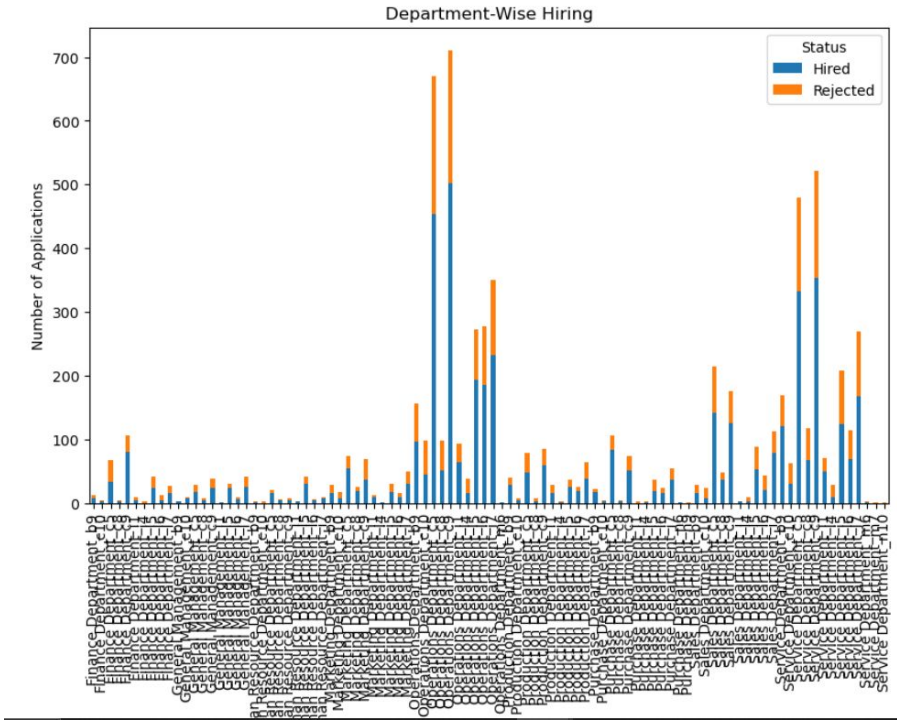
- Filter *Status* column to keep only the ***Hired*** values (plotted using matplotlib)
- Plot time series plot on ***Status*** column
- ***Interview Taken on*** is the time column
- We can see that the number of applications increased during August month, that is after the appraisal period of most companies, so applicants usually apply after graduation or appraisal.



Creating visualizations to better understand the data.

- Department wise hiring

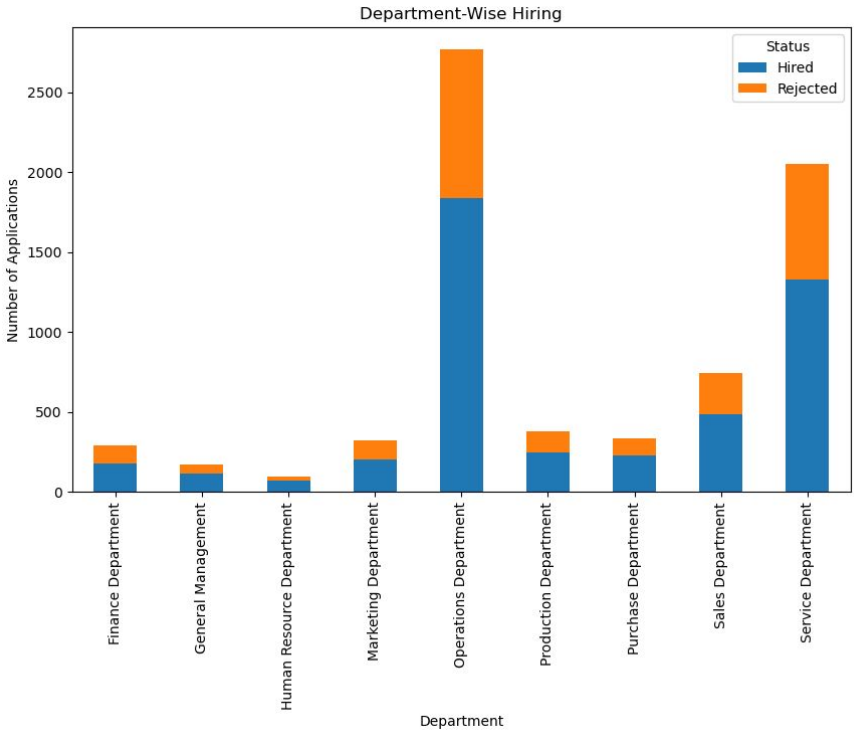
In this plot we can see the distribution of Department_postName column that we created wrt the Status column that is number of applicants hired or rejected for a particular post in a particular department.



Creating visualizations to better understand the data.

We can look at Department wise Hiring:

- Plot a column plot of people Hired and Rejected.
- The Departments Operations and Service have recruited more people in this cycle. And HR dept has recruited the least number of people

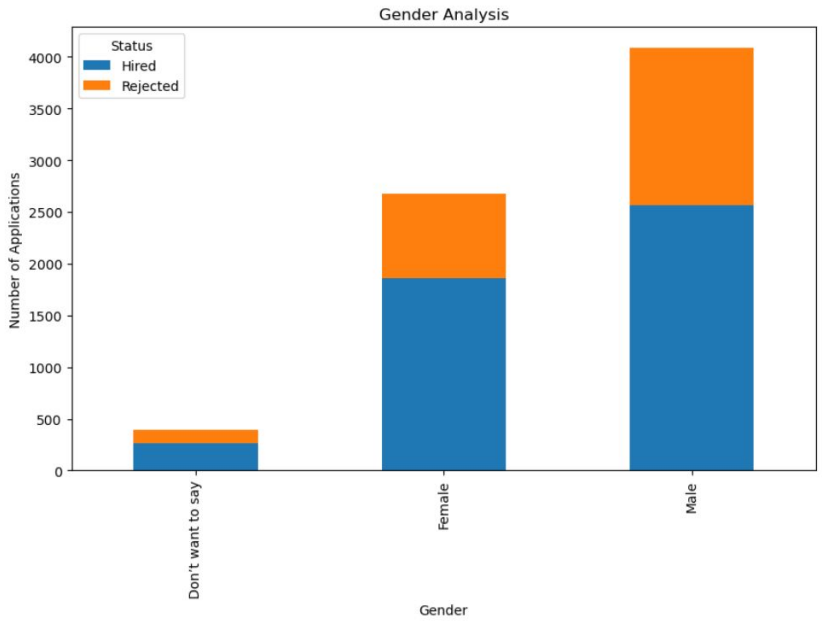


Creating visualizations to better understand the data.

- Gender based Analysis

We can see that the organization has a relatively higher number of Male applicants than Female ones

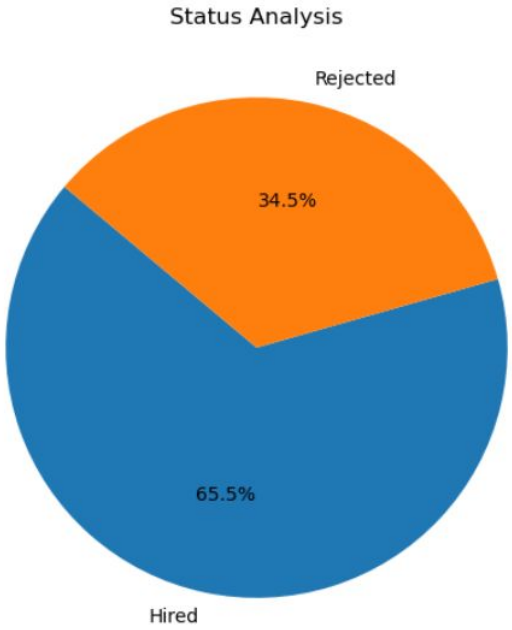
But the organization hires a equal distribution of people across all genders. The organization does not discriminate between Male and Female employee hires



Creating visualizations to better understand the data.

- Status Analysis

We can see that out of total applicants, almost 65.5% have the skills to get through all rounds and get hired.



Data Analytics Tasks

I. Hiring Analysis:

- Your Task: Determine the gender distribution of hires. How many males and females have been hired by the company?
- We can achieve this by using pivot tables to aggregate the data wrt Status and Event_name(Gender) columns

Steps:

1. Select pivot table on complete data
2. Take Status column as Filter column
3. Take Gender or event_name column as Rows column
4. Keep Count of Unique application id as the Values column

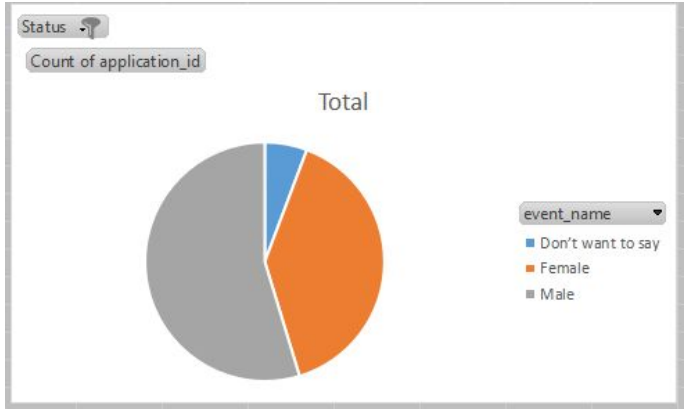
- The PivotTable displays the count of male and female hires.

	A	B	C
1	Status	Hired	
2			
3	Count of application_id		
4	event_name	Grand Total	
5	Don't want to say	268	
6	Female	1854	
7	Male	2561	
8	Grand Total	4683	
9			
10			
11			

I. Hiring Analysis: continued

Insights:

- The count of male is greater to that of female hires
- Male hire percentage is 54.6% and female hire percentage is almost 40%
- There are also few applicants about 6%, who do not want to disclose their gender



	percentage
Don't want to say	5.722827247
Female	39.59000641
Male	54.68716635

II. Salary Analysis:

- Your Task: What is the average salary offered by this company? Use Excel functions to calculate this.
- We can achieve this by using Average function in Excel on Offered Salary column
- To avoid discrepancies in the result I have removed the outliers during preprocessing

Steps:

1. Select a blank cell in Excel
2. Use =AVERAGE(G2, G7149) function to calculate average of the Offered Salary column
3. Enter
4. Alternatively can also sum the offered salary column and then divide it by the count of records in that column

-

III. Salary Distribution:

- Your Task: Create class intervals for the salaries in the company. This will help you understand the salary distribution.
- We can achieve this by using Frequency function in Excel on Offered Salary column

Steps:

1. Decide on the intervals and create new column
2. Take min and max using MIN(), MAX() functions
3. Take difference of Min and Max values as range of the data
4. Take a number i.e. 10 for dividing the data into intervals.
5. Take range divided by no of intervals for Interval difference

Average of Offered Salary	49881.14312
min	800
max	99967
total rows	7148
Range of data	99167
Number of intervals	10
Interval difference	9916.7

III. Salary Distribution: continued

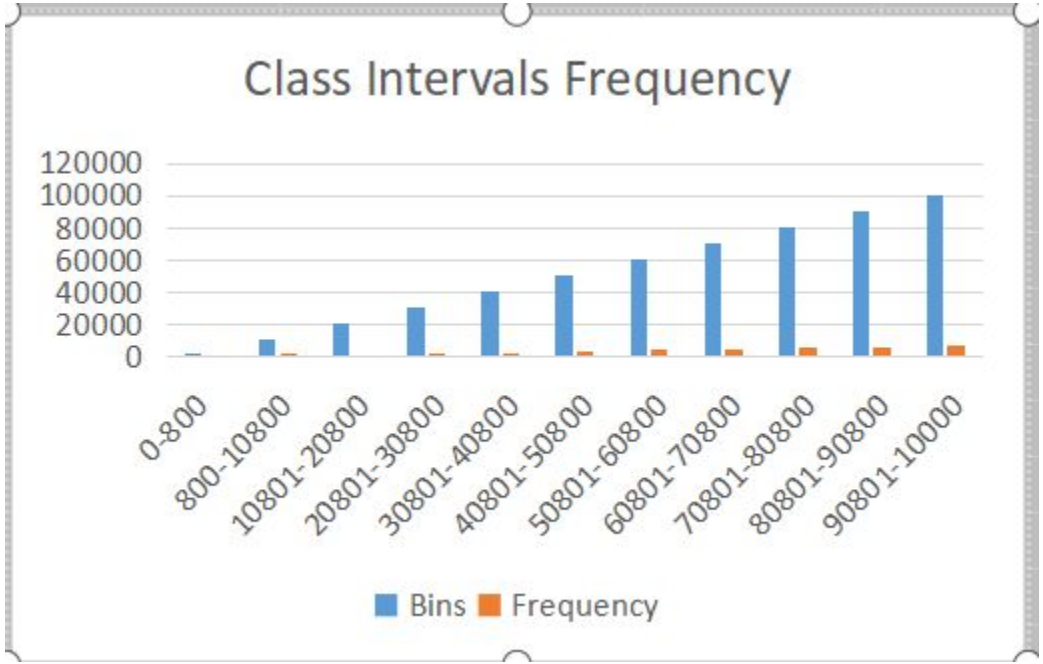
Steps continued:

- Then create a new column in data for making the intervals(bins) take formula *=J2+M12* for calculating subsequent intervals
- Use Frequency function to calculate frequencies =FREQUENCY(G2:G7149,J2:J12)

- This will give you the distribution of salaries across the intervals.

I	J	K
Class Intervals	**Bins**	**Frequency**
0-800	800	1
800-10800	10716.7	727
10801-20800	20633.4	1432
20801-30800	30550.1	2153
30801-40800	40466.8	2855
40801-50800	50383.5	3624
50801-60800	60300.2	4376
60801-70800	70216.9	5056
70801-80800	80133.6	5788
80801-90800	90050.3	6487
90801-10000	99967	7138

III. Salary Distribution: continued



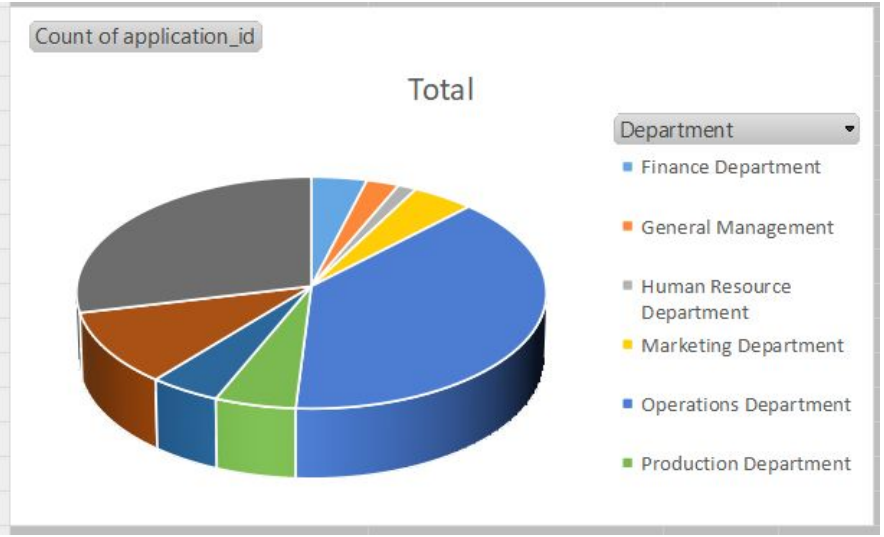
IV. Departmental Analysis:

- Your Task: Use a pie chart, bar graph, or any other suitable visualization to show the proportion of people working in different departments.
- We can achieve this by using pivot table to Aggregate data

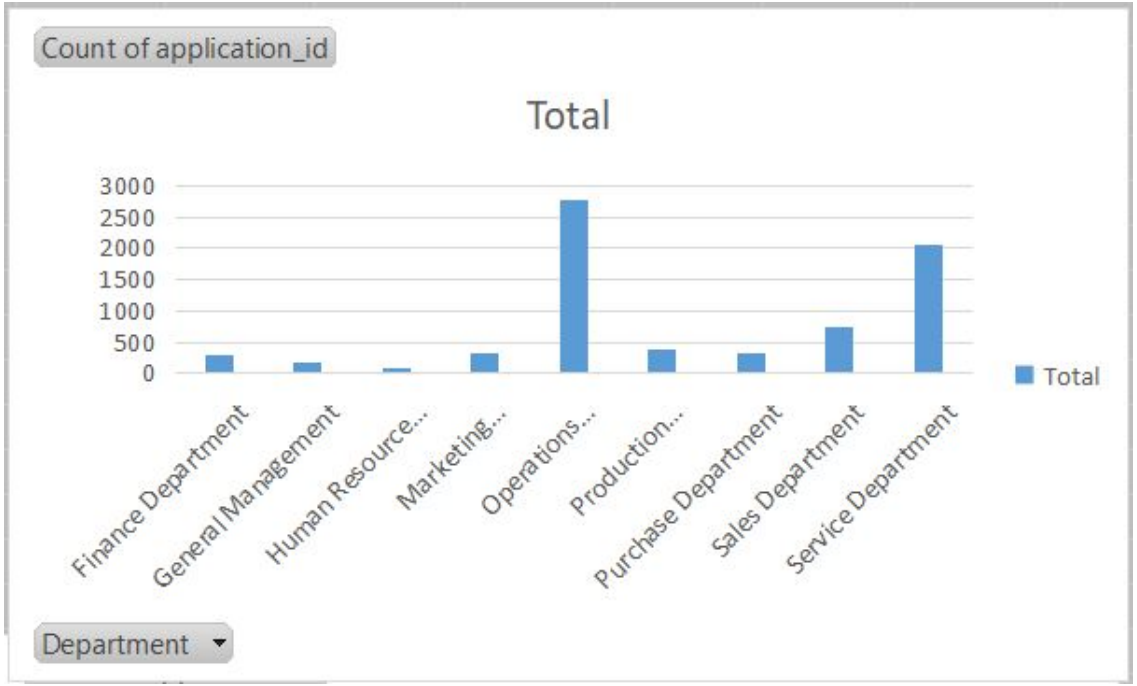
Steps

1. Keep Department as ***Rows***, and Count of Applicants in ***Values***
2. Add pie chart for count of values

This will visualize the proportion of people working in different departments.



IV. Departmental Analysis: continued



V. Position Tier Analysis:

- Your Task: Use a chart or graph to represent the different position tiers within the company. This will help you understand the distribution of positions across different tiers.
- We can achieve this by using pivot table to Aggregate data

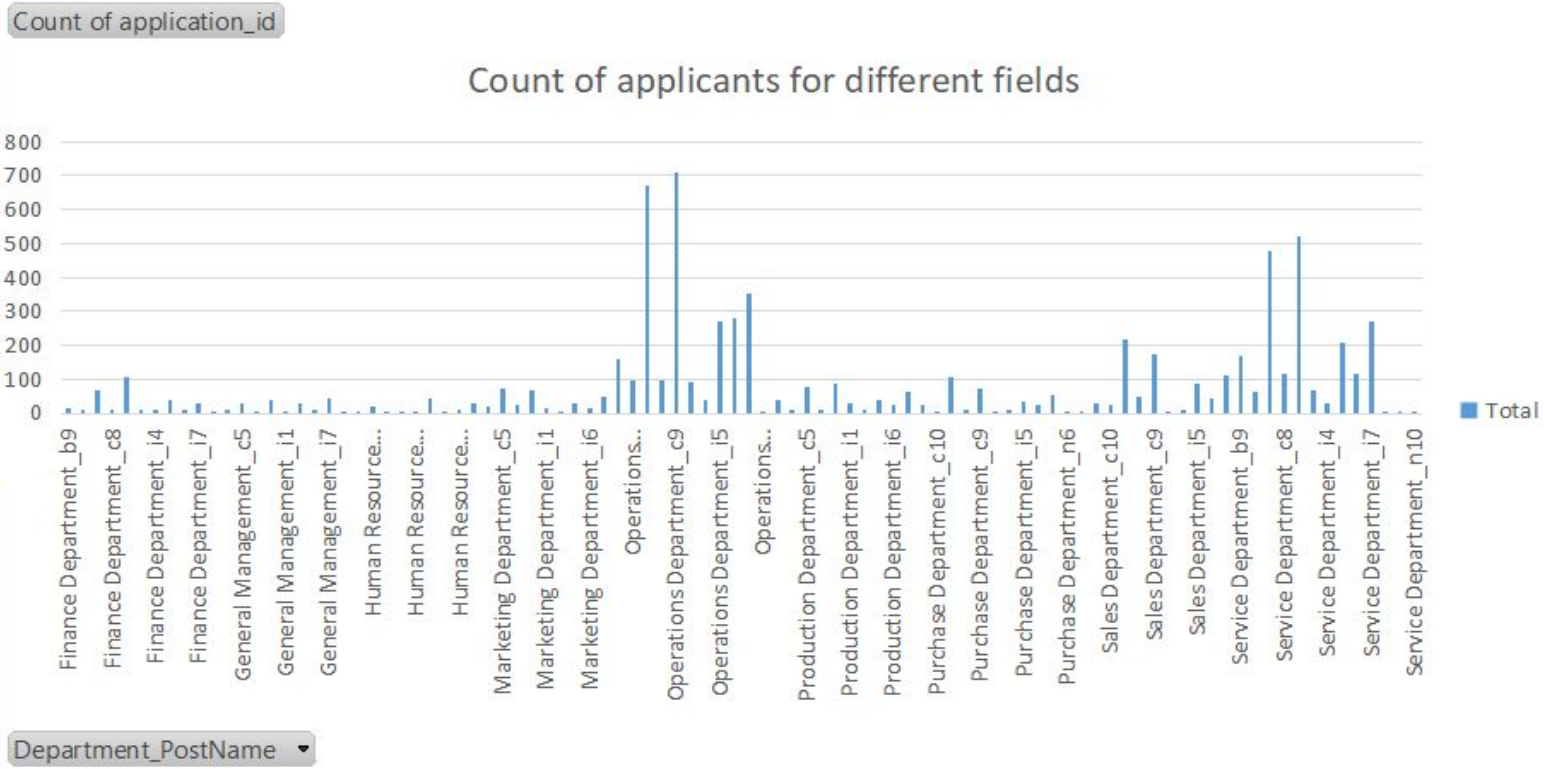
Steps

1. Keep *Department_PostName* as ***Rows***, and Count of Application ids in ***Values***
2. Add column chart for count of values

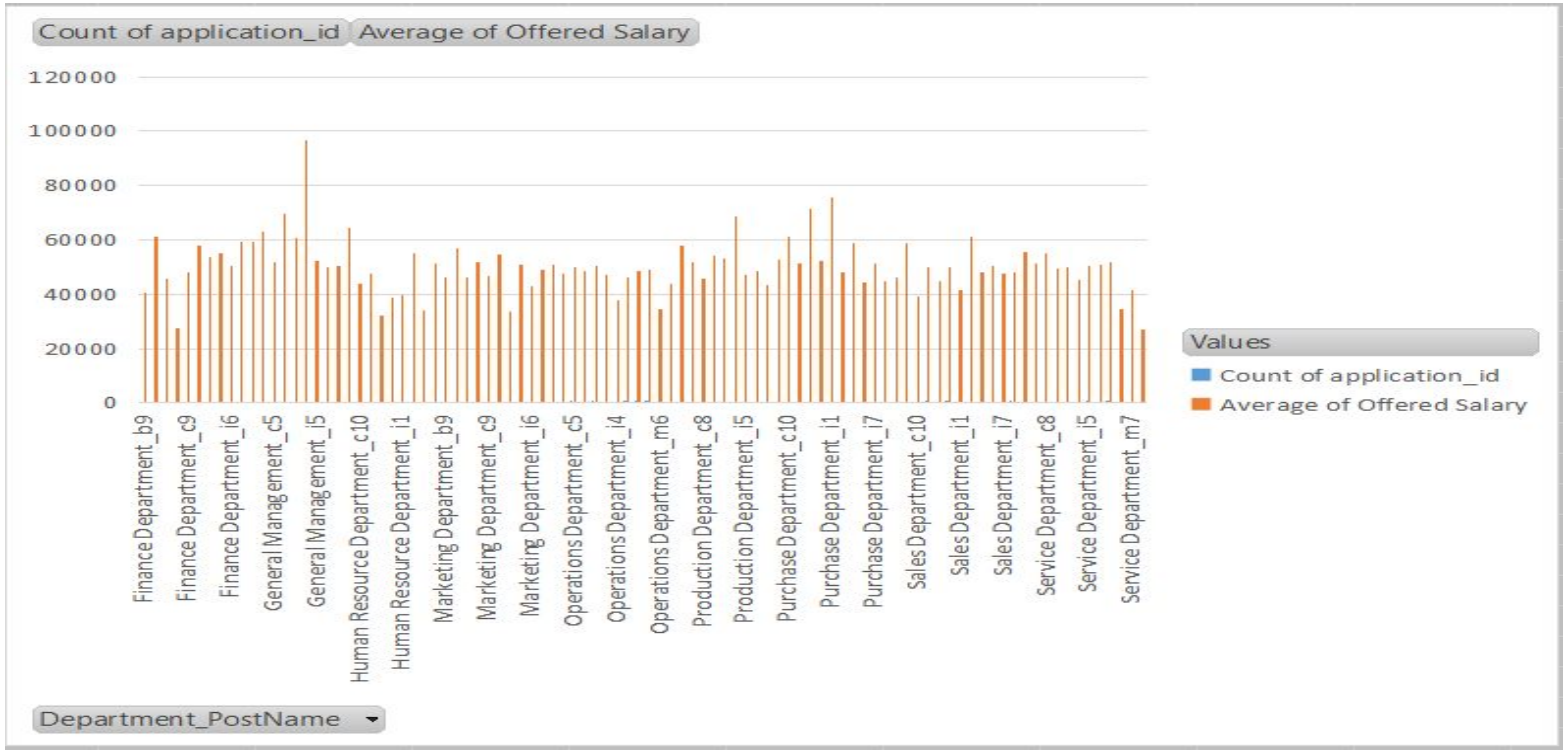
This will represent the distribution of positions across different tiers.

Department_PostName	Count of application_id
Finance Department_b9	13
Finance Department_c10	4
Finance Department_c5	68
Finance Department_c8	4
Finance Department_c9	107
Finance Department_i1	9
Finance Department_i4	3
Finance Department_i5	41
Finance Department_i6	12
Finance Department_i7	27
General Management_b9	2
General Management_c10	10
General Management_c5	29
General Management_c8	7
General Management_c9	39
General Management_i1	1
General Management_i5	31
General Management_i6	9
General Management_i7	42
Human Resource Department_b9	2
Human Resource Department_c10	2
Human Resource Department_c5	21
Human Resource Department_c8	6
Human Resource Department_c9	7
Human Resource Department_i1	2
Human Resource Department_i5	42
Human Resource Department_i6	6
Human Resource Department_i7	0

V. Position Tier Analysis: continued



V. Position Tier Analysis: continued



Project Report

Project Report

- **Project Description**
    - This project aims to analyze the hiring process data of a multinational company or organization
    - The objective is to draw meaningful insights from the data to help the hiring department improve its strategies and make better hiring decisions.
    - Key areas of focus include understanding gender distribution among hires, analyzing salary trends, examining departmental hiring patterns, and assessing the distribution of positions within the company.

Project Report

- **Approach**

After downloading the dataset and loading it into Excel, I started with the following steps:

1. Went through the complete data and saw that few values were missing i.e. data values were either blank or hyphenated So i filtered for these values and removed them
2. Preprocessed the data like combining two column to make a new column, etc
3. Attempted few visualizations in Excel and EDA using python to get to know the data more
4. Performed Hiring and Salary Analysis as per task lists by solving each problem step by step, took help from Online Videos wherever required.

Project Report

- **Tech-Stack Used**

1. WPS OFFICE version 12.2.0
2. Python ver 3.7.4 - ipynb files
3. Pandas, matplotlib libraries
4. VSCode ver 1.91.0

Jupyter-notebook link - https://drive.google.com/file/d/1njlgEJP0HH3VWUWPUv3Q_JSVMdULJ69t/view?usp=sharing

Project Report

- **Insights**

    1. The analysis revealed a balanced gender distribution among hires. There were no significant biases toward any gender, indicating that the company's hiring practices are equitable in terms of gender.
    2. The average salary offered by the company was calculated, providing a benchmark for evaluating the company's compensation structure. This helps in understanding whether the company offers competitive salaries compared to industry standards.
    3. The salary distribution analysis showed that most salaries fall within a specific range, with a few outliers at the higher end. This indicates that while the company generally offers competitive salaries, there are instances of significantly higher salaries, likely for senior or specialized roles.

Project Report

- **Insights continued**

    1. Outliers - The box plot identifies outliers in the salary data, which were mostly associated with high-salary positions. These outliers were expected and justified based on the roles and experience levels.
    2. The departmental analysis highlighted which departments have higher hiring rates and which face more rejections. Departments like Operations and Service had the highest hiring rates, indicating robust growth or high turnover in these areas.
    3. The position tier analysis revealed the distribution of hires across various levels, from entry-level to senior positions. This insight helps understand the company's hiring focus, whether it is more inclined towards bringing in fresh talent or hiring experienced professionals.

Project Report

- **Results**

1. The gender analysis showed a balanced hiring practice, suggesting the company is committed to diversity and inclusion
2. The average salary analysis indicated that the company offers competitive compensation, which is crucial for attracting and retaining top talent
3. The high hiring rates in departments like Operations and Engineering suggest a strategic focus on these areas, possibly due to business expansion or high project demands
4. Understanding the distribution across different position tiers helps in workforce planning and career progression mapping for employees.
5. Identifying outliers helps in understanding salary anomalies and ensuring they are justified by the role's requirements and market standards.

Project Report

- **Conclusion:**

The project provided valuable insights into the company's hiring process. Key findings highlighted a balanced gender distribution, competitive salaries, and a strategic focus on certain departments. The analysis also identified outliers in salary data, which are crucial for ensuring fair compensation practices. These insights can help the hiring department refine their strategies, improve hiring efficiency, and support the company's growth objectives.

Thank You

By Gayatri Palkar